

Iteration

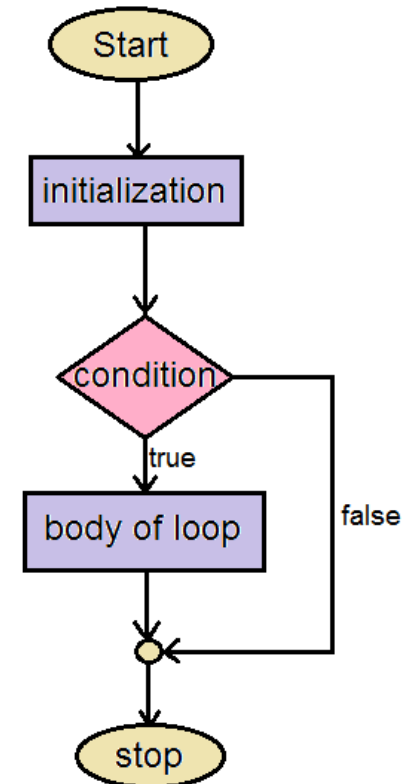
Introduction to computer programming

Management and Artificial Intelligence



Iteration

- **Iteration** is the process where a set of instructions (or statements) is executed repeatedly for a specified number of time or until a condition is met.
- Like branching, these statements also **alter the control flow** of the program.



Anatomy of a looping mechanism.

while loops

```
while <condition>:  
    statement  
    statement  
    ...  
    statement
```

- <condition> evaluates to a Boolean (True or False)
 1. if <condition> is True , then execute all statements inside the while block
 2. evaluate <condition> again
 3. repeat until <condition> is False

Example

```
%%script echo skipping  
  
n = input("You are in the Lost Forest. Go Left or Right?: ")  
while n == "Right":  
    n = input("You are in the Lost Forest. Go Left or Right?: ")  
print("You are out of the Lost Forest.")
```

skipping

for loops

```
for <variable> in range(<number>):  
    statement  
    statement  
    ...  
    statement
```

- each time through the loop, `<variable>` takes some value
- `<variable>` starts from the smallest value (usually, `1`)
- next time, `<variable> = <variable> + 1`

The `range()` function

The `range(start, stop, step)` function returns a sequence of numbers, starting from `start`, and increments by `step`, and stops at `stop-1`.

Keep in mind the following caveats:

- `stop` is mandatory
- `step` is optional: if not provided, it defaults to `1`
- `start` is optional: if not provided, it defaults to `0`

Examples:

```
mysum = 0
for i in range(7, 10):
    mysum = mysum + i
print(mysum)
```

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```
mysum = 0
for i in range(5, 11, 2):
    mysum = mysum + i
print(mysum)
```

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for vs. while loops

Iterate through sequence of numbers in `[0, 5)`:

- With a `while` loop:

```
n = 0
while n < 4:
    print(n)
    n = n + 1
```

```
0
1
2
3
```

- With a `for` loop:

```
for n in range(4):
    print(n)
```

```
0
1
2
3
```

Main difference

for loops

- **known** number of iterations
- can **end early** via `break`
- (natively) uses a **counter**
- you **can rewrite** a `for` loop using a `while` loop

while loops

- **unbounded** number of iterations
- can **end early** via `break`
- you can use a **counter**, but you must initialise it before the loop and increment it inside the loop
- you **may not be able to rewrite** a `while` loop using a `for` loop

More on `for` loops

`for` loops can be used to iterate over other data structures (e.g., lists or dictionaries*).

Example: print all functions in the `math` module that contain the word `'log'`;

```
import math
for f in dir(math):
    if 'log' in f:
        print(f)
print("Done!")
```

```
log
log10
log1p
log2
Done!
```

RECALL: `dir()` returns a list.

* We will see lists and dictionaries later in the course

Nested Loops

We know that branching (i.e., `if`) statements can be arbitrarily nested. The same holds for loops.

Example:

```
mysum = 0

for i in range(3):
    for x in range(2):
        mysum += i

print(mysum)
```

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RECALL: `mysum += i` is the same as `mysum = mysum + i`

break statement

The `break` statement:

- immediately exits whatever loop is it in
- all further statements in the block are skipped
- exits only the innermost loop
- usually triggered at given condition

```
while <condition_1>:
    while <condition_2>:
        <statement_a>
        break           # exits inner loop
        <statement_b>    # never executed
    <statement_c>
```

Example

```
mysum = 0

for i in range(5, 11, 2):
    mysum += i
    if mysum == 5:
        break
    mysum += i

print(mysum)
```

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What's going on here?

Nested loops with `break`, an example:

```
mysum = 0

for i in range(30):
    for n in range(20):
        if n > 2:
            break
        mysum += i
    if i > 1:
        break

print(mysum)
```

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What's going on here?

Exercises

Exercise 1: Coin flipping

- Ask the user how many coin tosses have to be performed
- Keep track of the number of times we get heads and tails
- We assume a *fair coin* (i.e., unbiased 50% chance of getting head and tail at each toss)

HINT: You may use the `random.randint()` function to simulate a single coin toss.

```
import random

tail = 0
head = 0

n = 3 # number of coin tosses
# n = int(input("How many coin tosses? "))
for i in range(n):
    x = random.randint(1, 100)
    if x > 50:
        head = head + 1
    else:
        tail = tail + 1

print("Number of heads: ", head)
print("Number of tails: ", tail)
```

Number of heads: 1
Number of tails: 2

Exercise 2: Sum of sequence

Write a function named `rolling_sum()` that:

- repeatedly asks the user to insert a number
- we assume integer numbers
- keeps track of the sum of the numbers inserted so far
- if the user inserts 0, then exit and return the sum

```
%%script echo skipping

def rolling_sum():
    this_number = int(input("Insert a number: "))
    my_sum = this_number

    while this_number != 0:
        this_number = int(input("Insert a number: "))
        my_sum = my_sum + this_number

    return my_sum

print("Total sum is: ", rolling_sum())
```

skipping

Exercise 3: Odd or even?

Write a function named `odd_or_even()` that:

- accepts an integer number, say `n`
- prints the number of odd numbers in range `[1,n]` (i.e., both included)
- prints the number of even numbers in range `[1,n]` (i.e., both included)

```
def odd_or_even(n):  
    odd = 0  
    even = 0  
  
    for i in range(1, n + 1):  
        if i % 2 == 0:  
            even += 1  
        else:  
            odd += 1  
  
    print("Number of odd elements: ", odd)  
    print("Number of even elements: ", even)  
  
odd_or_even(10)
```

Number of odd elements: 5
Number of even elements: 5

Exercise 4: The Fibonacci Sequence

Write a function named `fibonacci1()` that:

- accepts an integer number, say `n`
- prints the first `n` elements of the Fibonacci sequence.

Definition: Each number in the Fibonacci sequence is given by the sum of the two preceding ones. The Fibonacci sequence starts with two 'seed' numbers `0` and `1` and then the proper sequence starts: `0, 1, 1, 2, 3, 5, 8, 13, 21, ...`

```
def fibonacci1(n):
    second_to_last = 0
    last = 1

    print(second_to_last)
    print(last)

    for i in range(n - 2):
        new = last + second_to_last
        print(new)
        second_to_last = last
        last = new

fibonacci1(10)
```

0
1
1
2
3
5
8
13
21
34

Exercise 5: Multiples

Write a function named `multiples()` that:

- accepts an integer number, say `n`
- iterates the integers in `[1, n]`
- for multiples of 3, print `'Mult3'` instead of the number
- for multiples of 5, print `'Mult5'` instead of the number
- for multiples of 5 and 3, print `'Mult35'` instead of the number

```
def multiples(n):  
    for num in range(1, n + 1):  
        if num % 3 == 0 and num % 5 == 0:  
            print("Mult35")  
        elif num % 3 == 0:  
            print("Mult3")  
        elif num % 5 == 0:  
            print("Mult5")  
        else:  
            print(num)
```

```
multiples(15)
```

```
1  
2  
Mult3  
4  
Mult5  
Mult3  
7  
8  
Mult3  
Mult5  
11  
Mult3  
13
```


14
Mult35

Exercise 6: Print a subset

Write a function named `print_subset()` that:

- accepts an integer number, say n
- accepts an integer number, say $m \leq n$
- scans all integer numbers in range $[1, n]$, but prints all numbers in range $[1, m]$

Solution 1

```
def print_subset(n, m):  
    for num in range(1, n + 1):  
        if num <= m:  
            print(num)  
  
print_subset(5, 3)
```

1
2
3

Solution 2

```
def print_subset(n, m):  
    for num in range(1, n + 1):  
        print(num)  
        if num == m:  
            break  
  
print_subset(5, 3)
```

1
2
3

Stop and think: Under some circumstances, the second solution is more efficient than the first. What are those circumstances and why is it more efficient?

Exercise 7: Primality test

Write a function named `smallerPrimes()` that:

- accepts a positive integer number, say n
- prints all positive integer prime numbers smaller than n

Recall: A *prime* number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

Hint: The simplest primality test aims at checking whether n is evenly divisible (i.e., division with no remainder) by any number in range $[2, n/2]$. If so, the number is not prime.

```
def is_prime(num):  
    if num <= 1:  
        return False  
    for i in range(2, int(num/2) + 1):  
        if num % i == 0:  
            return False  
    return True  
  
def smallerPrimes(n):  
    for i in range(2, n):  
        if is_prime(i):  
            print(i)  
  
smallerPrimes(20)
```

2
3
5
7
11
13
17
19

Exercise 8: Pythagorean triples

Write a function named `pythagoreanTriples()` that:

- accepts an integer positive number, say n
- prints all x and y pairs such that $x^2 + y^2 = n^2$

Recall: The Pythagorean triple consists of three positive integers x , y and n , with $x, y \leq n$, such that: $x^2 + y^2 = n^2$

```
def pythagoreanTriples(n):  
    for x in range(1, n):  
        for y in range(1, n):  
            if x**2 + y**2 == n**2:  
                print(x, y)
```

```
pythagoreanTriples(5)
```

3 4

4 3

